Solution 1

a) The bending equation for the deflection $v(x)$ of a buckled beam that is pinned at both ends and subject to an axial end load is:

$$EIv'' = -Pv$$

where E is Young's modulus, I is the second moment of area of the beam, and P is the axial load. The general solution is of the form:

$$v = A\cos\kappa x + B\sin\kappa x \quad ; \quad \kappa = \sqrt{\frac{P}{EI}}$$

The boundary conditions are $v = 0, v'' = 0$ at $x = 0$ and $x = L$, from which it follows that $A = 0$ and $\kappa = n\pi/L$. The lowest order solution, where n = 1, is the only one that occurs in practice, and for this solution we have $k = \pi/L$. The end load on the buckled beam, which is also the critical load for buckling, is therefore:

$$P = EI\kappa^2 = \frac{\pi^2 EI}{L^2} \qquad [5]$$

b) The resonant frequency is given by $\omega_0 = \sqrt{\frac{k}{m}}$ and, from the *Information for Candidates*, the Q is given by $Q = \frac{m\omega_0 d}{A\mu}$. Under the described scenario, k and m scale as L^1 and L^3, so ω_0 will scale as L^{-1}. Also, since d is held constant, we have d and A scaling as L^0 and L^2 respectively, so the Q will scale as $\frac{L^3 L^{-1} L^0}{L^2} = L^0$ (invariant). [6]

c) For a D = 380 µm-thick wafer, the base of the trapezoidal etch pit should have a side length $W_b = 1000 + \sqrt{2} \times 380 = 1537.4$ µm. To compensate for the finite anisotropy, the mask aperture should be smaller than this by an amount $\delta = \frac{\sqrt{6}D}{S} = \sqrt{6} \times \frac{380}{35} = 26.6$ µm, so the required mask opening side length is $W_m = W_b - \delta = \mathbf{1510.8\ \mu m}$.

To guarantee the endpoint is reached, the etching process needs to run for:

$$T = \frac{D}{\min\{R_{100}\}} = 345 \text{ mins}$$

The largest possible undercut will then be $max\{\delta\} = \frac{\sqrt{6}max\{R_{111}\}T}{min\{S\}} = 36.6\ \mu m$. This is 10 µm larger than the nominal value, so the membrane will have a side length of **1010 µm**.

Conversely, the smallest undercut will be $min\{\delta\} = \frac{\sqrt{6}min\{R_{111}\}T}{max\{S\}} = 23.2\ \mu m$. This is 3.4 µm smaller than the nominal value, so the membrane will have a side length of **996.6 µm**. [6]

d) The piezoresistor will experience longitudinal and transverse stresses of σ_{max} and $\nu\sigma_{max}$ respectively and, from the *Information for Candidates*, we can see that this will result in a fractional resistance change of:

$$\frac{\Delta R}{R} = (\pi_L + \nu\pi_T)\sigma_{max}$$

where (also from the *Information for Candidates*):

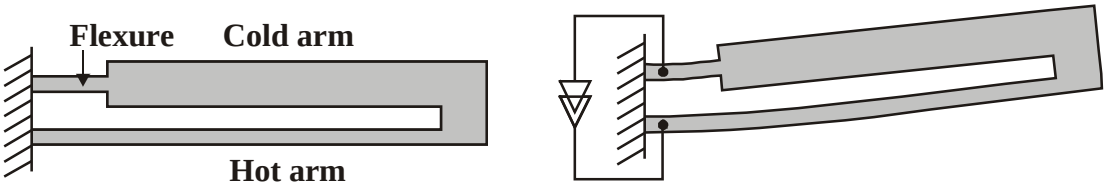
$$\sigma_{max} \approx 0.3\left(\frac{L}{h}\right)^2 p$$

From the definitions and numerical values provided, we have $\pi_L = 71.8 \times 10^{-11}$ Pa^{-1}, $\pi_T = -66.3 \times 10^{-11}$ Pa^{-1}, and $\upsilon = 0.064$. Substituting these values, along with $L = 2$ mm, $h = 10$ μm and $p = 100$ Pa, we find that $\frac{\Delta R}{R} = 8.1 \times 10^{-4}$ or 0.081%. [6]

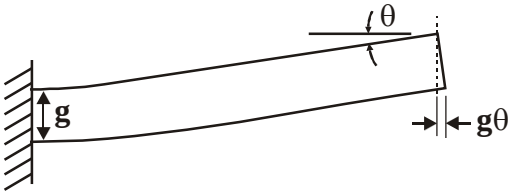
e) We need to evaluate $I = \int x^2 dA$ over the cross-section, and the quickest approach is to divide the cross-section into horizonal strips, where a given strip of height dy will contribute an amount $dI = \frac{1}{12} w^3 dy$ to the integral, with w being the local width. Now, we have $w = b - \alpha|y|$ where $\alpha = 2(b-a)/h$, so:

$$I = \frac{1}{6}\int_0^{h/2} (b - \alpha y)^3 \, dy = \frac{(b^4 - a^4)}{24\alpha} = \frac{(a^2+b^2)(a+b)h}{48}$$ [5]

f) Typical structure of an electrothermal actuator:



Current passed through structure causes differential heating because the cold arm has lower resistance (and hence less heat generated) per unit length, and better heat conduction. The structure deflects to allow differential expansion of the hot and cold arms. The difference in length between the hot and cold sides is approximately $\alpha(L_h T_h - L_c T_c - L_f T_f)$ where L_h, L_c and L_f are the lengths of the hot arm, cold arm and flexure respectively, and T_h, T_c and T_f are the corresponding average temp rises. This must be equal to $g\theta$, where θ is the (small) angular deflection and g is the gap between the beams:



Assuming the flexure bends as a circular arc, and the cold arm remains straight, the tip deflection is then:

$$v = (L_c + \tfrac{1}{2}L_f)\theta \approx \alpha(L_c + \tfrac{1}{2}L_f)(L_h T_h - L_c T_c - L_f T_f)/g$$ [6]

g) Comparison of electron-beam lithography (EBL) and focused ion beam (FIB):

Similarities: Both EBL and FIB use a focused beam of charged particles to modify a material locally. EBL and FIB systems are similar in construction, consisting of an electron/ion source, a vacuum column in which the electrons/ions are accelerated to high energy and then electrostatically focused and deflected, and a mechanical stage system. Both techniques are very slow and so not used in mass production of devices.

Differences: Electrons are several orders of magnitude lighter than ions and too light to cause physical damage; however, they can modify a material chemically by breaking bonds and this is the basis of EBL. In contrast, a beam of high-energy ions can be used to sputter material from a surface and therefore produce topological features directly. FIB can also be used to implant ions into a surface, permanently modifying its properties. The higher mass also makes ions easier to focus down to small spot size and less prone to scatter in the target material, and consequently the minimum feature size achievable with FIB is smaller than for EBL.

[6]

Solution 2

a) When looking for the normal modes for a wave equation such as the one given we assume a separable trial function with a harmonic time-varying part, i.e. a trial function of the form:

$$v(x,t) = X(x)\exp(j\omega t)$$

This function has the property that $\frac{\partial^2 v}{\partial t^2} = -\omega^2 v$, so if we substitute it into the wave equation the time-dependence will fall away and we will be left with the following spatial equation:

$$EI\frac{d^4X}{dx^4} = m\omega^2 X$$

for which the general solution is of the form:

$$X = A\cos kx + B\sin kx + C\cosh kx + D\sinh kx \qquad [6]$$

where k must satisfy the condition $\boldsymbol{EIk^4 = m\omega^2}$. [2]

b) The boundary conditions in the case of a beam that is built in at both ends are $v = 0$ and $\partial v/\partial x = 0$ at both $x = 0$ and $x = L$. Applying these to the general solution:

$$\begin{aligned} v = 0\ at\ x = 0\ &\Rightarrow A + C = 0\ ; \\ v' = 0\ at\ x = 0 &\Rightarrow B + D = 0\ ; \end{aligned} \qquad [4]$$

$$\begin{aligned} v = 0\ at\ x = L &\Rightarrow A(coskL - coshkL) + B(sinkL - sinhkL) = 0\ ; \\ v' = 0\ at\ x = L &\Rightarrow -A(sinkL + sinhkL) + B(coskL - coshkL) = 0 \end{aligned}$$

Eliminating A and B from the last two equations gives the required result. [4]

c) From part b), and the given root of the eigenvalue equation, we know that the frequency of the lowest mode is:

$$f_0 = \frac{1}{2\pi}\left(\frac{4.73}{L}\right)^2\sqrt{\frac{EI}{m}}$$

With E = 170 GPa, L = 500 μm, $I = wh^3/12 = 10 \times 3^3 \times 10^{-24}/12 = 2.25 \times 10^{-23}$ m^4, and $m = 10 \times (3 \times 2330 + 0.5 \times 1500) \times 10^{-12} = 7.74 \times 10^{-8}$ kg/m, this gives $\boldsymbol{f_0}$ **= 100 kHz**. [5]

Since $f_0 \propto \frac{1}{\sqrt{m}}$ we know that $\delta f_0/f_0 = -\frac{1}{2}\delta m/m$. Putting $\delta m = 1 \times 10^{-15}/L = 2 \times 10^{-12}$ kg/m, with m and f_0 as above, gives $\boldsymbol{\delta f_0}$ **= 1.3 Hz**. [3]

d) The maximum strain at any location along the beam will occur at the top and bottom surfaces and will be equal to $\varepsilon_{max} = \frac{h}{2}max\{v''\}$. We therefore need to evaluate v'' at $x = 0$ and at the peak of the oscillation cycle. Differentiating the general solution twice and putting $x = 0, t = 0$ we find:

$$max\{v''\} = v''(0,0) = (C - A)k^2 = -2Ak^2$$

We need to link A to the vibration amplitude, and to do this we use the relations derived in part a) to eliminate B, C and D from the expression for $v(x, 0)$. This gives:

$$v(x, 0) = A\{(\cos kx - \cosh kx) - 0.983(\sin kx - \sinh kx)\}$$

We can now relate A to the vibration amplitude v_0 at the centre of the beam, since:

$$v_0 = v\left(\tfrac{1}{2}L, 0\right) = -1.59A$$

Eliminating A from the expressions we have derived for $v''(0,0)$ and v_0, we obtain:

$$max\{v''\} = \frac{2k^2}{1.59}v_0 = \frac{2(4.73)^2}{1.59}\frac{v_0}{L^2} = 28\frac{v_0}{L^2}$$

from which the desired result follows. [6]

Solution 3

a) Using the parallel plate approximation, the capacitance between any given moving electrode and the fixed electrodes either side of it are:

$$C_{l/r} = \frac{\varepsilon_0(l+u_y)h}{(g \pm u_x)} \qquad [3]$$

where the + and – signs correspond respectively to the left and right capacitances, C_l and C_r, respectively. The electrostatic forces associated with these capacitors are given by the partial derivatives of the stored energy with respect to u_x or u_y:

$$F_{x,l/r} = \frac{\partial}{\partial u_x}\left\{\frac{\varepsilon_0(l+u_y)hV^2}{2(g \pm u_x)}\right\} = \mp\frac{\varepsilon_0(l+u_y)hV^2}{2(g \pm u_x)^2} \;;$$

$$F_{y,l/r} = \frac{\partial}{\partial u_y}\left\{\frac{\varepsilon_0(l+u_y)hV^2}{2(g \pm u_x)}\right\} = \frac{\varepsilon_0 hV^2}{2(g \pm u_x)} \qquad [5]$$

The total actuator force along *x* or *y* is then the sum of the contributions from the *N* electrode gaps. Taking account of the signs this gives:

$$F_x = \frac{N\varepsilon_0 hV^2(l+u_y)}{4}\left\{\frac{1}{(g-u_x)^2} - \frac{1}{(g+u_x)^2}\right\} = \frac{N\varepsilon_0 hV^2(l+u_y)}{4}\frac{4gu_x}{(g^2-u_x^2)^2} \;;$$

$$F_y = \frac{N\varepsilon_0 hV^2}{4}\left\{\frac{1}{(g-u_x)} + \frac{1}{(g+u_x)}\right\} = \frac{N\varepsilon_0 hV^2}{4}\frac{2g}{(g^2-u_x^2)} \qquad [2]$$

b) Taking moments for the upper flexure gives:

$$R_{1x}u_y - R_{1y}L - 2C_1 = 0 \qquad (1)$$

and similarly for the lower flexure:

$$R_{2x}u_y - R_{2y}L - 2C_2 = 0 \qquad (2)$$

[4]

Now considering the entire structure, resolving forces gives:

$$R_{1x} + R_{2x} + F_x = 0 \;; \qquad (3)$$

$$R_{1y} + R_{2y} + F_y = 0 \qquad (4)$$

Finally, taking moments about (0,0) gives:

$$R_{2x}\frac{S}{2} - R_{1x}\frac{S}{2} - R_{1y}\frac{L}{2} - R_{2y}\frac{L}{2} - C_1 - C_2 - F_x\frac{u_y}{2} = 0 \qquad (5)$$

where *S* is the width of the portal frame. With the elimination of C_1, C_2 and F_x from (5) by substitution from (1), (2) and (3), (5) reduces to:

$$R_{2x}\frac{S}{2} - R_{1x}\frac{S}{2} = 0$$

which, when combined with (3), implies that $R_{1x} = R_{2x} = -\frac{1}{2}F_x$ as required. [4]

c) From the *Information for Candidates*, deflection of the portal frame by u_y will result in an x-axis deflection of $u_x = -\frac{3u_y^2}{5L}$. This will result in an x-directed actuator force of:

$$F_x = -\frac{3N\varepsilon_0 hV^2(l+u_y)}{5L}\frac{gu_y^2}{(g^2-u_x^2)^2} \quad [6]$$

If we now use the equation for F_y to eliminate $N\varepsilon_0 hV^2$ from this equation, and we remove the quadratic terms in u_x, the desired result is obtained. [2]

d) From the result in part c), we can see that operating the actuator leads to compressive loading of the flexures. Given that the device is going unstable, presumably the compressive load is reaching the critical load for buckling.

The problem could be eliminated by offsetting the moving electrodes to the right as this will make the loading of the flexures less compressive / more tensile. [4]

Solution 4

a) From the information given we know that the approximate expression for k must be of the form $k \approx k_0\left(1+\frac{F}{F_0}\right)$, where k_0 is the transverse stiffness under zero axial load. We also know that $k_0 = \frac{12EI}{L^3}$, and this can be re-cast as $k_0 = \frac{12F_0}{\pi^2 L}$. Combining these we obtain the desired result. [4]

b) In any situation where the span of a flexure is constrained, lateral deflection of the flexure will result in a tensile axial load and this will increase the transverse stiffness. From the *Information for Candidates* we know that a lateral deflection u will be accompanied by a reduction of approximately $\frac{3u^2}{5L}$ in the span if the span is unconstrained. It follows that, if the span is fixed, the same lateral deflection will lead to a tensile axial strain of $\frac{3u^2}{5L^2}$. This will result in a tensile axial load of $F \approx \frac{3EAu^2}{5L^2}$ where A is the beam cross-section. The spring force due to the flexure will then be:

$$F_k = ku = k_0 u + \frac{3k_0 EA}{5L^2 F_0}u^3$$

We can see there is a third order non-linearity in this expression which causes the flexure to act as a non-linear hardening spring.

Non-linearity of this type is undesirable in resonant devices because it can lead to hysteresis and jumps in the frequence response. The most common suspension type that suffers from this issue is the hammock suspension. [6]

c) We can assume that the mass and the substrate expand as though unconstrained. This results in stretching of the flexures because the central portion of the mass expands more than the substrate between the flexure anchor points. [4]

Focusing on one side of the proof mass, if the ambient temperature rises by ΔT, expansion of the proof mass will force the distance AB between the flexure attachment points to increase by $\delta AB = (2L + L_s)\alpha\Delta T$, while the distance L_s between the anchors will increase by $\delta L_s = \alpha_s L_s \Delta T$. The length of each flexure will therefore increase by:

$$\delta L = \frac{\delta AB - \delta L_s}{2} = L\alpha\Delta T + \frac{1}{2}(\alpha - \alpha_s)L_s\Delta T$$

In the absence of a constraint, the flexure lengths would increase by $L\alpha\Delta T$, so the thermal strain resulting from CTE mismatch will be:

$$\Delta\varepsilon_{th} = \frac{(\alpha-\alpha_s)L_s\Delta T}{2L}$$

as required. [4]

d) The thermal strain will lead to a tensile axial load of $F = Ewh\Delta\varepsilon_{th}$, and we need the resulting fractional change in k to compensate for the fractional change due to the drop in Young's modulus, i.e. we require:

$$\frac{F}{F_0} = -\frac{1}{E}\frac{dE}{dT}\Delta T$$ [4]

Substituting for F and F_0 we obtain:

$$\frac{Ewh(\alpha-\alpha_s)L_s}{2L}\frac{12L^2}{\pi^2 Ewh^3} = -\frac{1}{E}\frac{dE}{dT}$$

which rearranges to give the desired result. [4]

e) With $h = 5\ \mu\text{m}, L = 200\ \mu\text{m}, \alpha = 13.3 \times 10^{-6}\ \text{K}^{-1}$, $\alpha_s = 2.5 \times 10^{-6}\ \text{K}^{-1}$, and $\frac{1}{E}\frac{dE}{dT} = -1.2 \times 10^{-3}\ \text{K}^{-1}$, the above equation gives $L_s = 22.8\ \mu\text{m}$. [4]

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